

## TEPLY KLYUCH, Russian Federation

WMO: 247710

Lat: 62.789N

Lon: 136.853E

Elev: 289

StdP: 97.91

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p) |
| 1             | -51.5      | -49.8 | -53.7                         | 0.0 | -51.2 | -52.2 | 0.0 | -49.3 | 2.7                   | -31.1 | 2.5 | -34.4 | 0.9                   | 70   | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.7                   | 28.5            | 17.6 | 26.5 | 16.3 | 24.5 | 15.9 | 18.8                | 25.3 | 17.9 | 24.1 | 16.9 | 22.5 | 2.5                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.7                            | 12.3 | 20.2 | 15.7 | 11.5 | 19.7 | 14.5 | 10.7 | 19.2 | 54.9          | 25.1 | 51.7 | 24.1 | 48.6 | 22.4 | 24.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 5.3           | 4.4  | 3.7 | -53.0                      | 30.9 | 1.3                | 0.9 | -53.9                                              | 31.6 | -54.7      | 32.2 | -55.4      | 32.7 | -56.3      | 33.4 |    |    |
| (5)               |      |     | -53.0                      | 20.7 | 1.2                | 1.5 | -53.9                                              | 21.8 | -54.6      | 22.6 | -55.3      | 23.4 | -56.2      | 24.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -10.6  | -41.0 | -35.1 | -21.6 | -6.9  | 5.4   | 14.1  | 16.6  | 13.3  | 4.6   | -9.7  | -29.7 | -38.8 |  |  |
|                                                                        | DBStd       | 21.70  | 5.69  | 5.45  | 8.04  | 6.18  | 4.75  | 3.50  | 3.45  | 3.56  | 4.56  | 6.98  | 8.21  | 7.30  |  |  |
|                                                                        | HDD10.0     | 7989   | 1582  | 1263  | 980   | 507   | 153   | 7     | 2     | 11    | 167   | 612   | 1191  | 1514  |  |  |
|                                                                        | HDD18.3     | 10592  | 1840  | 1497  | 1238  | 757   | 401   | 132   | 74    | 159   | 411   | 870   | 1441  | 1772  |  |  |
|                                                                        | CDD10.0     | 466    | 0     | 0     | 0     | 0     | 11    | 129   | 206   | 113   | 7     | 0     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 28     | 0     | 0     | 0     | 0     | 0     | 4     | 21    | 3     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 641    | 0     | 0     | 0     | 0     | 3     | 171   | 377   | 89    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 142    | 0     | 0     | 0     | 0     | 0     | 32    | 97    | 13    | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.5    | 0.9   | 1.0   | 1.4   | 1.9   | 2.2   | 2.1   | 1.8   | 1.7   | 1.7   | 1.5   | 1.2   | 1.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 371    | 9     | 6     | 8     | 16    | 38    | 55    | 73    | 57    | 49    | 34    | 18    | 8     |  |  |
|                                                                        | PrecMax     | 495    | 20    | 18    | 27    | 47    | 72    | 103   | 181   | 95    | 91    | 61    | 53    | 23    |  |  |
|                                                                        | PrecMin     | 292    | 2     | 2     | 3     | 4     | 16    | 27    | 29    | 29    | 11    | 10    | 4     | 1     |  |  |
|                                                                        | PrecStd     | 52     | 4     | 4     | 5     | 10    | 13    | 20    | 33    | 17    | 21    | 12    | 9     | 5     |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | -18.6 | -14.3 | 0.7   | 9.7   | 23.1  | 29.6  | 31.3  | 28.3  | 21.2  | 6.9   | -0.5  | -10.0 |  |  |
|                                                                        |             | MCWB   | -18.9 | -14.8 | -2.8  | 3.5   | 11.2  | 17.4  | 18.7  | 18.0  | 12.6  | 3.1   | -1.0  | -10.3 |  |  |
|                                                                        | 2%          | DB     | -23.5 | -18.7 | -1.6  | 7.4   | 19.9  | 27.5  | 29.4  | 26.1  | 18.2  | 3.4   | -8.1  | -17.0 |  |  |
|                                                                        |             | MCWB   | -23.7 | -19.1 | -4.2  | 1.8   | 9.6   | 15.4  | 18.1  | 16.6  | 10.9  | 1.3   | -8.6  | -17.1 |  |  |
|                                                                        | 5%          | DB     | -28.5 | -22.7 | -3.7  | 5.4   | 17.5  | 25.4  | 27.5  | 23.8  | 15.9  | 1.5   | -13.1 | -23.2 |  |  |
|                                                                        |             | MCWB   | -28.7 | -23.1 | -6.0  | 0.4   | 8.6   | 14.6  | 17.6  | 16.0  | 9.7   | 0.1   | -13.6 | -23.4 |  |  |
|                                                                        | 10%         | DB     | -32.6 | -26.1 | -6.4  | 3.5   | 14.8  | 23.3  | 25.4  | 21.7  | 13.0  | -0.1  | -17.7 | -28.6 |  |  |
|                                                                        |             | MCWB   | -32.7 | -26.4 | -8.3  | -0.7  | 7.4   | 14.0  | 17.0  | 15.0  | 8.0   | -1.1  | -18.1 | -28.7 |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -18.7 | -14.8 | -1.8  | 3.7   | 11.9  | 18.5  | 21.2  | 19.2  | 13.1  | 3.9   | -1.0  | -10.5 |  |  |
|                                                                        |             | MCDB   | -18.3 | -14.3 | -0.3  | 9.4   | 22.3  | 26.6  | 26.8  | 24.7  | 20.2  | 5.8   | -0.4  | -9.8  |  |  |
|                                                                        | 2%          | WB     | -23.7 | -19.1 | -4.2  | 2.0   | 10.1  | 17.1  | 19.3  | 18.1  | 11.7  | 1.8   | -8.5  | -17.5 |  |  |
|                                                                        |             | MCDB   | -23.4 | -18.7 | -1.6  | 6.3   | 18.5  | 24.9  | 26.1  | 23.3  | 16.4  | 3.1   | -8.1  | -17.3 |  |  |
|                                                                        | 5%          | WB     | -28.6 | -23.1 | -6.1  | 0.8   | 9.1   | 16.0  | 18.5  | 16.9  | 10.4  | 0.2   | -13.5 | -23.5 |  |  |
|                                                                        |             | MCDB   | -28.5 | -22.7 | -3.8  | 4.8   | 16.6  | 22.9  | 25.0  | 22.0  | 14.5  | 1.2   | -13.1 | -23.2 |  |  |
|                                                                        | 10%         | WB     | -32.7 | -26.3 | -8.4  | -0.4  | 7.9   | 15.0  | 17.6  | 15.9  | 8.8   | -1.4  | -18.0 | -28.8 |  |  |
|                                                                        |             | MCDB   | -32.6 | -26.1 | -6.4  | 3.1   | 13.4  | 21.3  | 23.7  | 20.4  | 12.2  | -0.1  | -17.6 | -28.7 |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.3   | 14.6  | 19.4  | 15.8  | 11.7  | 13.6  | 12.7  | 13.1  | 11.5  | 9.8   | 9.8   | 7.1   |  |  |
|                                                                        |             | MCDBR  | 12.3  | 17.3  | 17.9  | 14.4  | 17.5  | 19.1  | 17.6  | 17.1  | 16.8  | 10.4  | 11.9  | 13.0  |  |  |
|                                                                        |             | MCWBR  | 12.2  | 16.6  | 14.9  | 9.6   | 8.9   | 8.7   | 7.7   | 8.7   | 9.5   | 7.9   | 11.4  | 12.3  |  |  |
|                                                                        | 5% WB       | MCDBR  | 12.3  | 17.1  | 16.9  | 13.0  | 15.8  | 15.0  | 14.3  | 14.0  | 14.1  | 8.9   | 12.0  | 12.4  |  |  |
|                                                                        |             | MCWBR  | 12.1  | 16.5  | 14.3  | 8.8   | 8.2   | 7.5   | 7.1   | 7.9   | 8.6   | 6.8   | 11.4  | 12.1  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.188 | 0.243 | 0.294 | 0.365 | 0.414 | 0.405 | 0.490 | 0.422 | 0.337 | 0.270 | 0.197 | 0.166 |  |  |
|                                                                        | taud        |        | 1.979 | 2.113 | 2.070 | 1.962 | 2.097 | 2.285 | 2.023 | 2.201 | 2.407 | 2.246 | 1.971 | 1.901 |  |  |
|                                                                        | Ebn at Noon |        | 482   | 711   | 794   | 798   | 794   | 815   | 719   | 743   | 761   | 668   | 439   | 288   |  |  |
|                                                                        | Edh at Noon |        | 33    | 65    | 103   | 143   | 138   | 117   | 149   | 114   | 76    | 57    | 31    | 17    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.30  | 1.28  | 3.13  | 5.11  | 5.22  | 5.44  | 5.09  | 3.98  | 2.24  | 1.18  | 0.47  | 0.14  |  |  |
|                                                                        | RadStd      |        | 0.02  | 0.06  | 0.12  | 0.20  | 0.32  | 0.50  | 0.39  | 0.34  | 0.31  | 0.11  | 0.04  | 0.01  |  |  |

## Historical Trends

|                             |     | DBAvg |     | Heating |        | Cooling |       | Degree-Days |         |         |         |
|-----------------------------|-----|-------|-----|---------|--------|---------|-------|-------------|---------|---------|---------|
|                             |     |       |     | 99% DB  | 99% DP | 1% DB   | 1% WB | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                         | (e) | (f)   | (g) | (h)     | (i)    | (j)     | (k)   | (l)         | (m)     |         |         |
| (46) Station Only           | N/A | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (3 neighbors) | N/A | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page